

Homework 1(a) - Frequency Distribution

1. Data Collection:

- Collect a dataset of at least 30 observations based on a topic of your interest (e.g., hours of study, scores in a test, number of messages sent in a day, etc.).
- Clearly describe the variable and its relevance to the context of your dataset.

Solution: The dataset chosen consists of the number of hours studied by 30 students and their corresponding test scores. The variable *hours studied* is a quantitative variable that represents the independent variable, while *test scores* is the dependent variable.

2. Data Classification:

- Classify the data as *qualitative* or *quantitative*.
- If quantitative, specify whether the data is *discrete* or *continuous*.

Solution: The data collected is *quantitative* because it consists of numerical values. The variable *hours studied* is *continuous*, as it can take any value within a range, whereas *test scores* are *discrete*, as they are typically whole numbers.